Rossmoyne Senior High School

Semester One Examination, 2019

Question/Answer booklet

MATHEMATICS METHODS UNIT 1

Section Two: Calculator-assumed

If required by your examination administrator, please place your student identification label in this box

Student number: In figures

In words____

Circle your Teacher's Surname: Benko Bestall Fraser-Jones Goh Koulianos Murray Rudland

Time allowed for this section

Reading time before commencing work: ten minutes
Working time: one hundred minutes

Materials required/recommended for this section

To be provided by the supervisor
This Question/Answer booklet
Formula sheet (retained from Section One)

To be provided by the candidate
Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: drawing instruments, templates, notes on two unfolded sheets of A4 paper, and up to three calculators approved for use in this examination

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	8	8	50	50	35
Section Two: Calculator-assumed	13	13	100	100	65
				Total	100

Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.

2. Write your answers in this Question/Answer booklet preferably using a blue/black pen. Do not use erasable or gel pens.

3. You must be careful to confine your answer to the specific question asked and to follow any instructions that are specified to a particular question.

4. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.

5. It is recommended that you do not use pencil, except in diagrams.

6. Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.

7. The Formula sheet is not to be handed in with your Question/Answer booklet.

Section Two: Calculator-assumed **65% (100 Marks)**

This section has **thirteen (13)** questions. Answer **all** questions. Write your answers in the spaces provided.

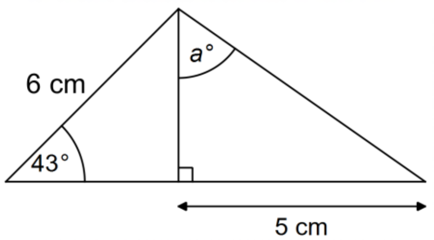
Working time: 100 minutes.

Question 9 **(9 marks)**

(a) The points A and B have coordinates $(4, 5)$ and $(-5, 12)$ respectively. If A is the midpoint of B and C, determine the coordinates of C. (3 marks)

(b) The points D and E have coordinates $(-3p, 2q)$ and $(4q, -p)$ respectively, where p and q are constants. Determine the value of p and the value of q if the midpoint of D and E is at $(-18, 7)$. (3 marks)

(c) Determine the value of *a*, correct to 2 decimal places. (3 marks)

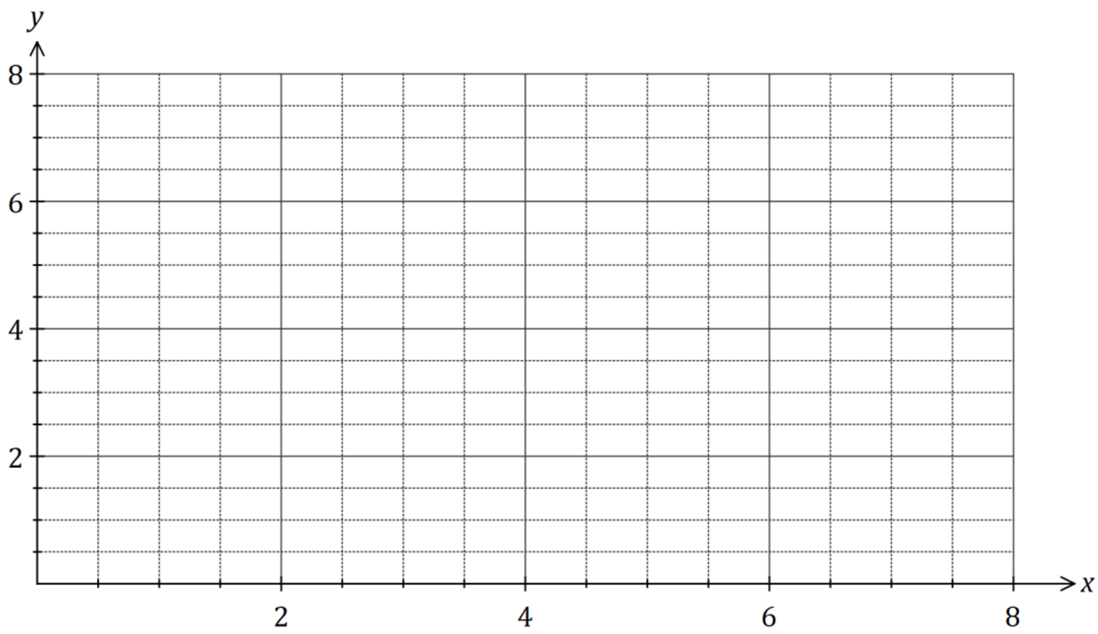


Question 10 **(9 marks)**

In an experiment, the signal intensity, S, can be modelled by $S(x) = 7.7 - 6x + 2.1x^2 - 0.2x^3$, where x is the distance from the signal source in metres and $0 \le x \le 7$.

(a) Determine S when $x = 6$. (1 mark)

(b) Draw the graph of $y = S(x)$ on the axes below showing the co-ordinates of any turning points or intercepts. (4 marks)



(c) Determine the equation of the straight line L that passes through the x-intercept and the y-intercept of the graph of $y = S(x)$. (2 marks)

(d) Determine the coordinates of the point of intersection of L with the graph of $y = S(x)$ where $x > 0$ and $y > 0$. (2 mark)

Question 11 **(9 marks)**

A positive integer **less than** 12 is chosen at random.

The outcome sets for events A, B and C are $A = \{3, 4, 5, 6\}$, $B = \{1, 3, 5, 7\}$ and $C = \{2, 6, 10\}$.

(a) List the following sets:

(i) $A \cap B$. (1 mark)

(ii) $A \cup B \cup C$. (2 marks)

(iii) $A \cap (B \cup C)$. (2 mark)

(b) Determine

(i) $\mathrm{n}(B \cap C')$. (1 mark)

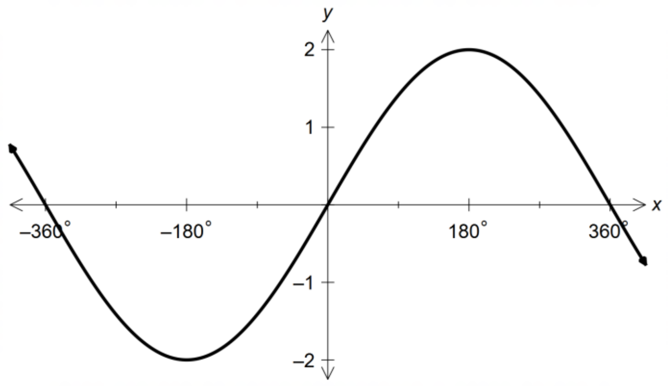
(ii) $P(B \cap \mathrm{C})$. (1 mark)

(iii) $P\left(A' \middle| (B \cup C)\right)$. (2 marks)

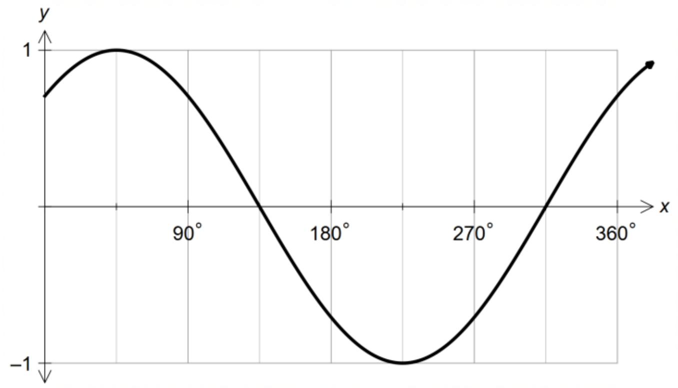
Question 12 **(6 marks)**

Determine the equations of each of the following trigonometric graphs.

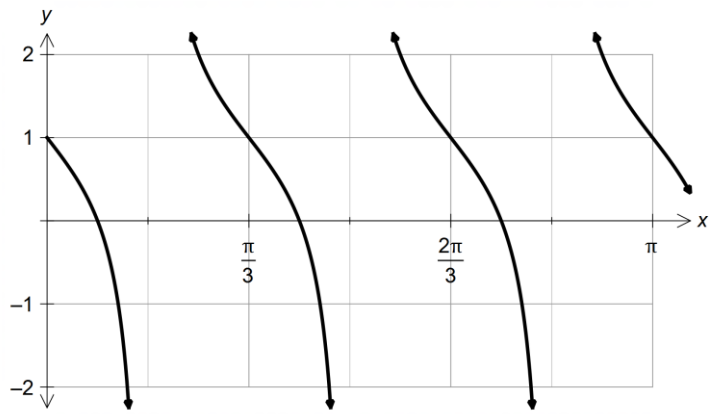
(a) (2 marks)



(b) (2 marks)



(c) (2 marks)



Question 13 **(7 marks)**

(a) Convert, giving an **exact** answer

(i) $16°$ to radians. (1 mark)

(ii) 0.4 radians to degrees. (1 mark)

(b) Calculate, to the nearest degree, the acute angle between the line $y = 1.5x - 4$ and the x-axis. (2 marks)

(c) The sides adjacent to the right-angle in a right triangle have lengths 65 cm and 72 cm.

If the smallest angle in the triangle is α, determine an exact value for

(i) $\tan \alpha$. (1 mark)

(ii) $\sin(90° - \alpha)$. (2 marks)

Question 14 **(7 marks)**

An **obtuse** angled triangle PQR has $p = 31$ cm, $r = 38$ cm and an area of 525 cm^2.

(a) Sketch a triangle to show this information. (1 mark)

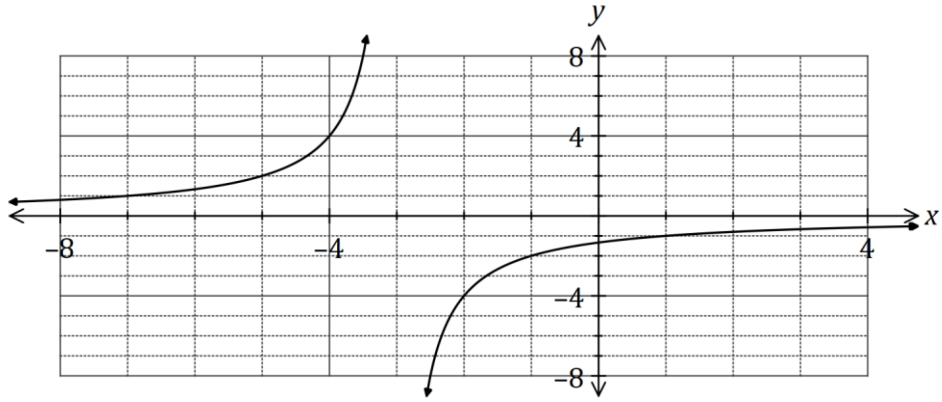
(b) Determine the size of $\angle Q$. (2 marks)

(c) Show that $q \approx 59$ cm. (2 marks)

(d) Show that $\angle P \approx 28°$. (2 marks)

Question 15 **(8 marks)**

The graph of $y = f(x)$ is shown below where $f(x) = \dfrac{a}{b - x}$.



(a) State the value of the constant a and the value of the constant b. (3 marks)

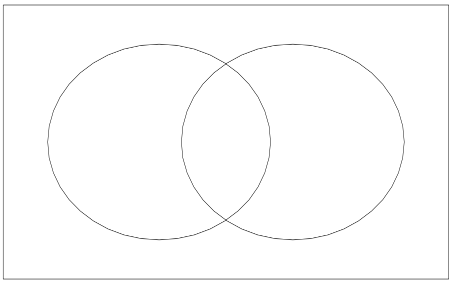
(b) The hyperbola shown above has two asymptotes. State their equations. (2 marks)

(c) Describe how to transform the graph of $y = f(x)$ to obtain the graph of $y = f(x) + 1$ and state the domain and range of the transformed function. (3 marks)

Question 16 **(7 marks)**

An examination consisted of two papers, one of which was much harder than the other. 12% of candidates gained a distinction in the first paper and 4% gained a distinction in the second paper whilst 87% did not gain a distinction in either paper.

(a) Represent the above information on the venn diagram below by placing a percentage in each region. (2 marks)



(b) Determine the probability that a randomly chosen candidate

(i) gained a distinction in both papers. (1 marks)

(ii) gained a distinction in one paper but not the other. (2 marks)

(iii) gained a distinction in the second paper given that they gained a distinction in the first. (2 mark)

Question 17 **(8 marks)**

(a) The variables P and w are directly proportional and when $w = 30, P = 6$.

(i) Determine an equation for the relationship between P and w. (2 marks)

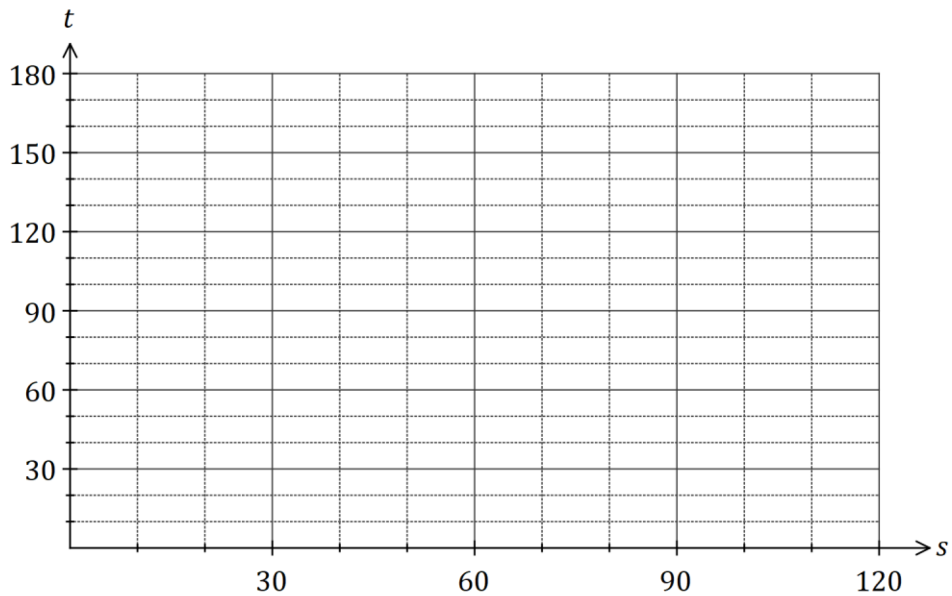
(ii) State the value of P when $w = 10$. (1 mark)

(b) The time, t minutes, that a car takes to travel 500 m at a constant speed of s kmh^{-1} is given by the formula $t = \frac{k}{s}$.

(i) Determine the value of the constant k, given that when $s = 30, t = 60$. (1 mark)

(ii) Determine the value of t when $s = 20$. (1 mark)

(iii) On the axes below, draw a graph to show how s varies with t. (3 marks)

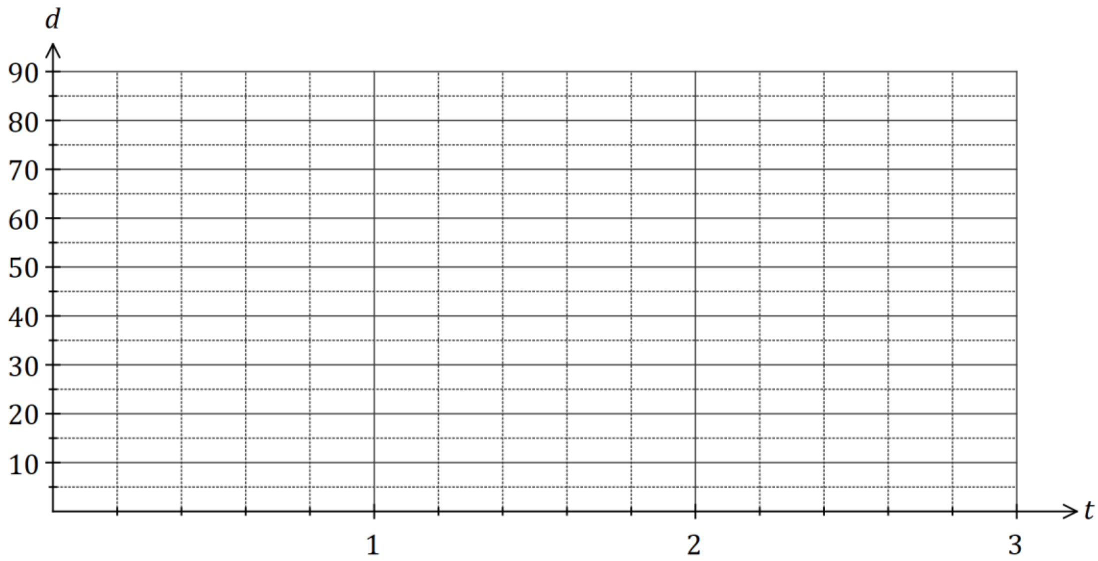


Question 18 **(8 marks)**

A small weight, attached to the bottom of a spring, oscillated up and down. The distance, d cm, of the weight from the top of the spring after t seconds can be modelled by

$$d = 45 + 35\sin\left(\frac{3\pi t}{4}\right)$$

(a) Sketch the graph on the axes below for $0 \le t \le 3$. (4 marks)



(b) Mark on your graph point M, where the weight is 40 cm from the top of the spring and moving downwards. (1 mark)

(c) Determine

(i) the maximum distance of the weight from the top of the spring. (1 mark)

(ii) the time taken for the weight to first return to its initial position. (1 mark)

(iii) the distance moved by the weight between $t = 1$ and $t = 2$. (1 mark)

Question 19 **(8 marks)**

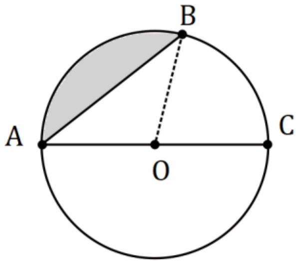
(a) The equation of the axis of symmetry for the graph of $y = 2x^2 + 8x + 5$ is $x = m$. Determine the value of m, using a method that does not refer to the graph of the parabola. (2 marks)

(b) A parabola with equation $y = ax^2 + bx + c$ has a turning point at $(4, -5)$ and passes through the point $(2, -17)$. Determine the value of a, the value of b and the value of c. (3 marks)

(c) Determine the value of the discriminant for the quadratic equation $4x^2 - 28x + 47 = 0$ and use it to explain how many solutions the equation $(x + 3)(4x^2 - 28x + 47) = 0$ will have. (3 marks)

Question 20 **(8 marks)**

(a) The circle shown has centre O and diameter AC of length 80 cm. Determine the shaded area given that $3 \times \angle AOB = 5 \times \angle BOC$. (4 marks)



(b) A sector of a circle has a perimeter of 84 cm and an area of 360 cm^2. Determine the radius of the circle. (4 marks)

Question 21 **(6 marks)**

Let $a = \sin 50°$ and $b = \cos 100°$.

Give your answers to the following in terms of a and/or b.

(a) Write down an expression for

(i) $\sin 130°$. (1 mark)

(ii) $\cos 80°$. (1 mark)

(b) Determine an expression for $\cos 130°$. (3 marks)

(c) Determine an expression for $\tan 130°$. (1 mark)

End of Questions

Supplementary page

Question number: __________

Supplementary page

Question number: _________

Supplementary page

Question number: _________